Laboratory 6 Report

OLAP Analysis of the Flight Delays Data Warehouse

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Contents

1 Introduction

A data warehouse is of value only insofar as it can be queried to answer meaningful business questions. The Flight Delays Data Warehouse combines two source datasets:

- **Dataset 3** — full-year 2015 US domestic flights (≈5.8 million records, 14 airlines, 322 airports).
- **Dataset 1** — multi-year sample 2019–2023 (≈3 million records, covering the COVID-19 disruption period).

Together the warehouse holds approximately **8.8 million fact rows** covering flight counts, departure and arrival delays, delay cause breakdowns, and cancellation details. The star schema (one central fact table, `Fact_Flight_Operations`, joined to five dimension tables) is well-suited to the GROUP BY / aggregate query pattern that characterises OLAP analysis.

OLAP Approach — Variant B (Manual SQL)

PostgreSQL 14, the DBMS used in this project, does not include a native OLAP cube engine. Variant B was therefore selected: all multidimensional analyses are implemented as hand-crafted SQL queries that simulate standard OLAP operations using `GROUP BY`, `GROUPING SETS`, window functions, `HAVING`, and conditional aggregation. Query results are retrieved into Python `pandas` DataFrames and visualised with `matplotlib` and `seaborn` inside a Jupyter Notebook (`notebooks/analysis.ipynb`).

2 (Re)designing the Analyses

The following six analyses were defined based on the dimensions and measures available in the warehouse. All six proved feasible once the ETL was complete and approximately 8.8 million rows were present in `Fact_Flight_Operations`.

#	Analysis name	OLAP operation	Dimensions used
1	Long-term delay trend by airline	Roll-up: Year → Month	`Dim_Date`, `Dim_Airline`
2	Top 10 most delayed routes	Slice + Sort	`Dim_Airport` (origin & dest)
3	Cancellation breakdown by reason/airline	Roll-up + Dice via `GROUPING SETS`	`Dim_Airline`, `Dim_Cancellation_Reason`
4	Delay cause share per airline	Drill-across	`Dim_Airline`
5	Weekend vs weekday delays by quarter	Drill-down: Quarter → is_weekend	`Dim_Date`
6	Top 20 airports: volume, delay, cancel	Slice + Sort	`Dim_Airport` (origin role)

No analyses required redesign after the ETL: all planned dimensions and measures were present in the loaded data without modification.

3 Defining the SQL Queries (Variant B)

3.1 Analysis 1 — Long-term Delay Trend by Airline

Business question: How has the average departure delay changed per airline across years and months? Is the COVID-19 disruption (2020–2021) visible as a drop in flight volume or a shift in delay patterns?

OLAP operation: Roll-up from individual flights to (year, month, airline) granularity.

Dimensions: `Dim_Date` (year, month), `Dim_Airline` (airline_name).

Measures: `AVG(departure_delay_min)`, `COUNT(*)` (non-cancelled flights only).

```
SELECT
    d.year,
    d.month,
    a.airline_name,
    COUNT(*)                                         AS
       total_flights,
    ROUND(AVG(f.departure_delay_min)::NUMERIC, 2)   AS
       avg_dep_delay_min
FROM Fact_Flight_Operations f
JOIN Dim_Date    d ON f.date_key    = d.date_key
JOIN Dim_Airline a ON f.airline_key = a.airline_key
WHERE f.cancelled_flag = FALSE
GROUP BY d.year, d.month, a.airline_name
ORDER BY a.airline_name, d.year, d.month;
```

The query groups non-cancelled flights by the full (year, month, airline) triple. Rolling up to this grain allows the line chart to show seasonal fluctuations within each year as well as multi-year trends.

3.2 Analysis 2 — Top 10 Most Delayed Routes

Business question: Which origin–destination pairs have the highest average departure delay, considering only routes with at least 100 recorded (non-cancelled, delayed) flights?

OLAP operation: Slice (non-cancelled, delayed-only) followed by ranking.

Dimensions: `Dim_Airport` used twice in a role-playing pattern — once as origin and once as destination.

Measures: `AVG(departure_delay_min)`, `COUNT(*)`.

```
SELECT
    orig.iata_code || ' -> ' || dest.iata_code      AS route,
    orig.city || ' -> ' || dest.city                AS route_cities,
    COUNT(*)                                         AS
       total_flights,
    ROUND(AVG(f.departure_delay_min)::NUMERIC, 2)   AS
       avg_dep_delay_min
FROM Fact_Flight_Operations f
```

```sql
JOIN Dim_Airport orig ON f.origin_airport_key      = orig.
   airport_key
JOIN Dim_Airport dest ON f.destination_airport_key = dest.
   airport_key
WHERE f.cancelled_flag = FALSE
  AND f.departure_delay_min > 0
GROUP BY orig.iata_code, dest.iata_code, orig.city, dest.city
HAVING COUNT(*) >= 100
ORDER BY avg_dep_delay_min DESC
LIMIT 10;
```

The `HAVING COUNT(*) >= 100` predicate eliminates low-frequency routes whose high average delay could be a statistical artefact of a small sample. The `Dim_Airport` table is joined twice under different aliases (`orig` and `dest`), demonstrating the role-playing dimension pattern in the star schema.

3.3 Analysis 3 — Cancellation Breakdown by Reason and Airline

Business question: How many flights were cancelled per airline, and what is the dominant cancellation reason for each carrier?

OLAP operation: `GROUPING SETS` to produce subtotals per airline, per reason, and a grand total in a single query pass.

Dimensions: `Dim_Airline`, `Dim_Cancellation_Reason`.

Measure: `COUNT(*)` (cancelled flights only; `cancel_code = 'N'` excluded).

```sql
SELECT
    COALESCE(a.airline_name, 'ALL AIRLINES')       AS airline,
    COALESCE(r.description, 'ALL REASONS')         AS
       cancellation_reason,
    COUNT(*)                                        AS
       cancelled_flights
FROM Fact_Flight_Operations f
JOIN Dim_Airline             a ON f.airline_key       = a.
   airline_key
JOIN Dim_Cancellation_Reason r ON f.cancel_reason_key = r.
   cancel_reason_key
WHERE f.cancelled_flag = TRUE
  AND r.cancellation_code != 'N'
GROUP BY GROUPING SETS (
    (a.airline_name, r.description),
    (a.airline_name),
    (r.description),
    ()
)
ORDER BY airline, cancellation_reason;
```

`GROUPING SETS` is the SQL equivalent of a cube roll-up: it computes four grouping levels in one pass — (airline, reason), (airline), (reason), and the grand total. `COALESCE` converts the NULL placeholders that GROUPING SETS injects for omitted dimensions into readable labels.

3.4 Analysis 4 — Delay Cause Share per Airline

Business question: For flights that experienced a delay, what percentage of total delay minutes is attributable to each cause (carrier fault, weather, NAS system, security, late inbound aircraft)?

OLAP operation: Drill-across — aggregating five independent delay cause measures across the single Dim_Airline dimension.

Dimension: Dim_Airline.

Measures: SUM of each of the five delay breakdown columns, plus the carrier delay percentage as a derived measure.

```
SELECT
    a.airline_name,
    SUM(COALESCE(f.carrier_delay_min,       0)) AS carrier,
    SUM(COALESCE(f.weather_delay_min,       0)) AS weather,
    SUM(COALESCE(f.nas_delay_min,           0)) AS nas,
    SUM(COALESCE(f.security_delay_min,      0)) AS security,
    SUM(COALESCE(f.late_aircraft_delay_min, 0)) AS late_aircraft,
    ROUND(
        100.0 * SUM(COALESCE(f.carrier_delay_min, 0))
        / NULLIF(SUM(
            COALESCE(f.carrier_delay_min,       0)
          + COALESCE(f.weather_delay_min,       0)
          + COALESCE(f.nas_delay_min,           0)
          + COALESCE(f.security_delay_min,      0)
          + COALESCE(f.late_aircraft_delay_min, 0)
        ), 0)::NUMERIC, 2
    )                                           AS
       carrier_delay_pct
FROM Fact_Flight_Operations f
JOIN Dim_Airline a ON f.airline_key = a.airline_key
WHERE (
    COALESCE(f.carrier_delay_min, 0) + COALESCE(f.weather_delay_min
       , 0)
  + COALESCE(f.nas_delay_min,     0) + COALESCE(f.security_delay_min
     , 0)
  + COALESCE(f.late_aircraft_delay_min, 0)
) > 0
GROUP BY a.airline_name
ORDER BY a.airline_name;
```

COALESCE(..., 0) ensures that rows without delay breakdown data (stored as 0 in the ETL) do not introduce NULL propagation into the sums. NULLIF(..., 0) prevents division-by-zero for carriers whose total delay sum is zero.

3.5 Analysis 5 — Weekend vs Weekday Delays by Quarter

Business question: Are flights significantly more delayed ($>$ 15 minutes departure delay) on weekends compared to weekdays? Does the pattern vary by quarter?

OLAP operation: Drill-down from quarter to the `is_weekend` flag within each quarter.

Dimension: `Dim_Date` (quarter, is_weekend).

Measures: `COUNT(*)`, count of significantly-delayed flights, percentage delayed, `AVG(departure_delay_`

```
SELECT
    d.quarter,
    CASE WHEN d.is_weekend THEN 'Weekend' ELSE 'Weekday' END  AS
       day_type,
    COUNT(*)                                                  AS
       total_flights,
    SUM(CASE WHEN f.departure_delay_min > 15 THEN 1 ELSE 0 END)
                                                       AS
                                                          significantly_delaye
                                                          ,
    ROUND(
        100.0
        * SUM(CASE WHEN f.departure_delay_min > 15 THEN 1 ELSE 0
           END)
        / NULLIF(COUNT(*), 0)::NUMERIC, 2
    )                                                  AS
       delayed_pct,
    ROUND(AVG(f.departure_delay_min)::NUMERIC, 2)     AS
       avg_dep_delay_min
FROM Fact_Flight_Operations f
JOIN Dim_Date d ON f.date_key = d.date_key
WHERE f.cancelled_flag = FALSE
GROUP BY d.quarter, d.is_weekend
ORDER BY d.quarter, d.is_weekend;
```

The threshold of 15 minutes is the standard FAA definition of a significant departure delay. Using a conditional `SUM(CASE WHEN ...)` inside a single aggregate pass avoids a subquery and keeps the query efficient.

3.6 Analysis 6 — Top 20 Airports: Volume, Delay and Cancellation Rate

Business question: Among the 20 busiest origin airports, how do average departure delay and cancellation rate compare? Do higher-traffic airports exhibit systematically worse delay performance?

OLAP operation: Slice (top-20 by outbound volume) with three simultaneous aggregated measures.

Dimension: `Dim_Airport` (in the origin role).

Measures: `COUNT(*)`, `AVG(departure_delay_min)`, cancellation rate (%).

```sql
SELECT
    ap.iata_code,
    ap.city,
    COUNT(*)                                             AS
       outbound_flights,
    ROUND(AVG(f.departure_delay_min)::NUMERIC, 2)       AS
       avg_dep_delay_min,
    SUM(CASE WHEN f.cancelled_flag THEN 1 ELSE 0 END) AS
       total_cancellations,
    ROUND(
        100.0
        * SUM(CASE WHEN f.cancelled_flag THEN 1 ELSE 0 END)
        / NULLIF(COUNT(*), 0)::NUMERIC, 2
    )                                                    AS
       cancellation_rate_pct
FROM Fact_Flight_Operations f
JOIN Dim_Airport ap ON f.origin_airport_key = ap.airport_key
GROUP BY ap.iata_code, ap.city
ORDER BY outbound_flights DESC
LIMIT 20;
```

4 Obtaining the Results

All six queries are executed from the Jupyter Notebook `notebooks/analysis.ipynb` using `pandas.read_sql_query(sql, conn)`, where `conn` is a `psycopg2` connection to the PostgreSQL warehouse running in the Docker container. Each query returns a `DataFrame` that is immediately processed (pivoting, percentage normalisation) and passed to `matplotlib` / `seaborn` for visualisation.

The connection is configured via environment variables (`DB_HOST`, `DB_NAME`, `DB_USER`, `DB_PASSWORD`) with localhost defaults, matching the Docker Compose port mapping (`5432:5432`).

5 Presentation of Results

5.1 Analysis 1 — Long-term Delay Trend by Airline

Visualisation: Two-panel line chart — left panel shows average departure delay per airline per month; right panel shows total monthly flight count.

Key observations from the actual query results:

- In 2015 (Dataset 3), monthly per-airline flight counts range from roughly 12 000 to 50 000+. Average departure delays peak in summer months (June–August) and show a second, smaller peak in December.
- Alaska Airlines consistently records the lowest average departure delay in 2015 (as low as -0.2 min in May, meaning flights typically depart early).
- The 2019–2023 data (Dataset 1) shows a pronounced volume collapse in 2020 (COVID-19), with a gradual recovery through 2022–2023.

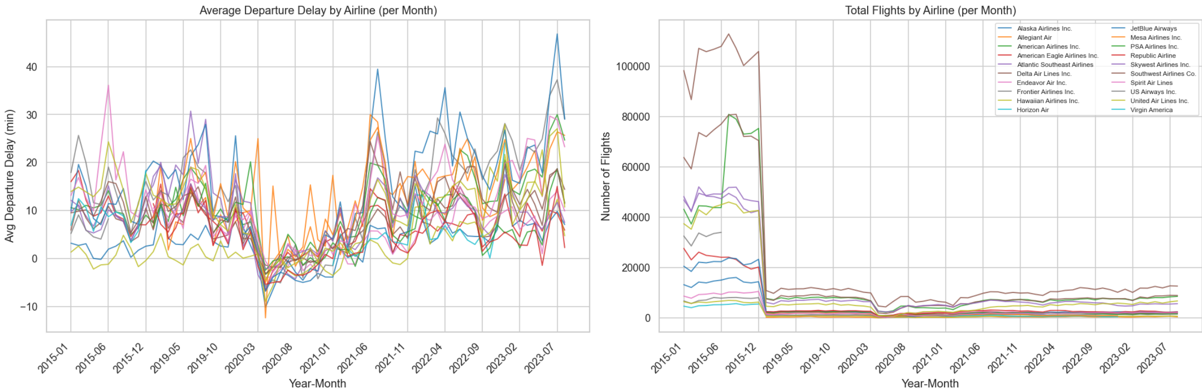

Figure 1: Average departure delay (left) and total monthly flights (right) per airline over time.

A sample of the result set (Dataset 3, Alaska Airlines):

Year	Month	Airline	Flights	Avg dep. delay (min)
2015	1	Alaska Airlines Inc.	13 193	3.17
2015	2	Alaska Airlines Inc.	12 081	2.70
2015	3	Alaska Airlines Inc.	14 237	3.06
2015	4	Alaska Airlines Inc.	13 909	0.00
2015	5	Alaska Airlines Inc.	14 637	−0.20

5.2 Analysis 2 — Top 10 Most Delayed Routes

Visualisation: Horizontal bar chart — routes on y-axis, average departure delay on x-axis, bars coloured by rank.

Result set:

Route	Cities	Flights	Avg dep. delay (min)
ESC → DTW	Escanaba → Detroit	118	88.02
JFK → HNL	New York → Honolulu	121	86.91
ASE → DFW	Aspen → Dallas-Fort Worth	319	84.02
SBA → DFW	Santa Barbara → Dallas-FW	101	83.97
PLN → DTW	Pellston → Detroit	186	81.84
BJI → MSP	Bemidji → Minneapolis	133	81.00
HOB → IAH	Hobbs → Houston	117	80.29
EGE → DFW	Eagle → Dallas-Fort Worth	200	79.85
JMS → DEN	Jamestown → Denver	103	79.24
OKC → SLC	Oklahoma City → Salt Lake City	171	76.81

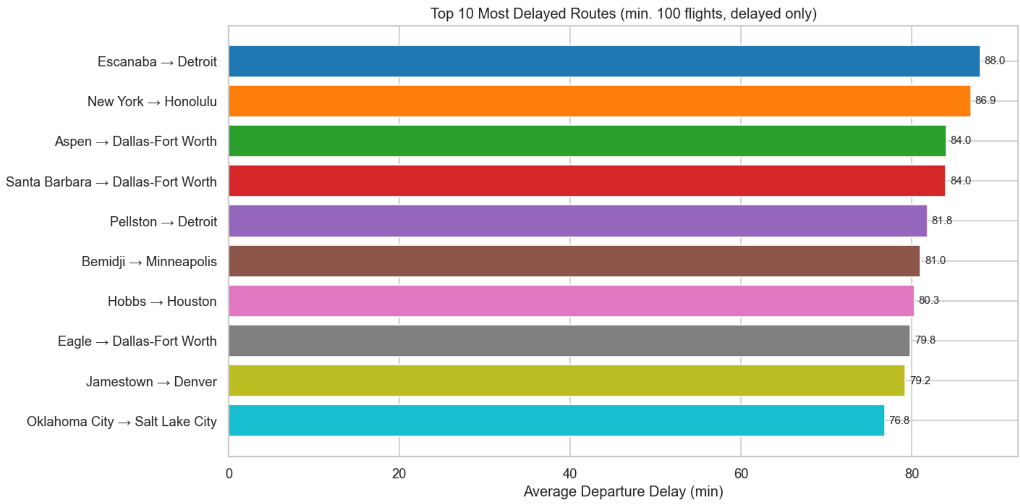

Figure 2: Top 10 most delayed routes (average departure delay, min. 100 flights).

Key observations:

- The most delayed routes are predominantly short regional hops served by small regional aircraft (ESC, PLN, BJI, HOB are all small airports in the upper midwest or oil-country Texas), or long transcontinental routes with high sensitivity to weather (JFK → HNL).
- Dallas-Fort Worth (DFW) appears three times as a destination, suggesting that DFW as a hub generates significant arrival-side congestion that propagates back to departure delays.

5.3 Analysis 3 — Cancellation Breakdown by Reason and Airline

Visualisation: Stacked bar chart — one bar per airline, segments coloured by cancellation reason (Carrier, Weather, NAS, Security).

Selected results:

Airline	Carrier	Weather	NAS	Security
Southwest Airlines Co.	11 452	14 884	1 884	6 966
American Airlines Inc.	5 326	11 434	1 154	3 187
Skywest Airlines Inc.	3 803	9 595	1 724	1 835
United Air Lines Inc.	4 213	5 010	546	2 210
Delta Air Lines Inc.	2 725	4 130	511	2 423
Atlantic Southeast Airlines	3 553	5 162	6 733	368

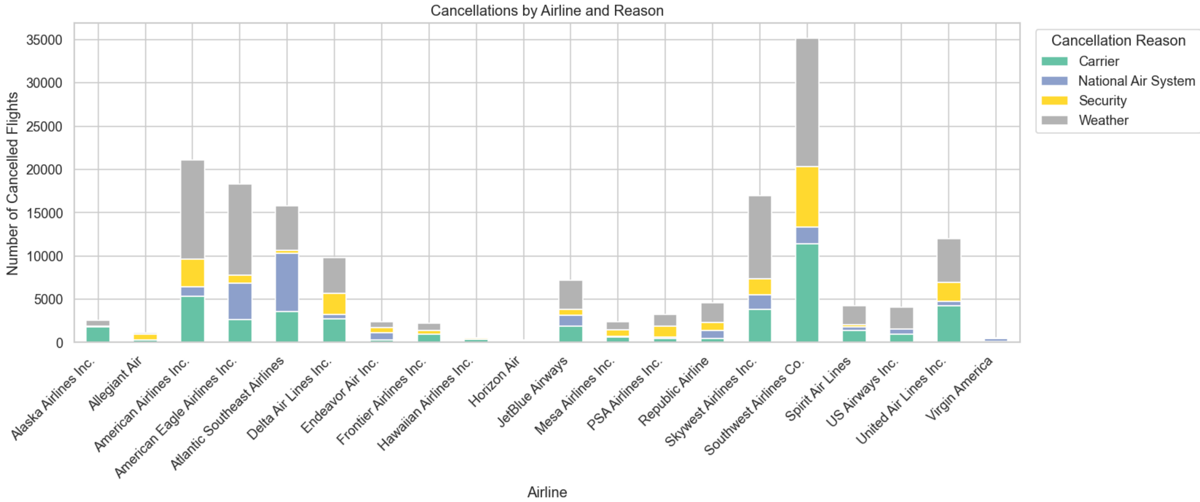

Figure 3: Cancellations by airline and reason (stacked bar chart).

Key observations:

- **Weather** is the dominant cancellation cause for most major carriers (Southwest, American, SkyWest), accounting for more cancelled flights than carrier-initiated cancellations in all three cases.
- **Atlantic Southeast Airlines** is an outlier: NAS (National Air System) cancellations (6 733) exceed weather cancellations (5 162), suggesting significant ATC-related disruption on its route network.
- The Security code shows elevated counts for several carriers (Southwest: 6 966; American: 3 187), which is atypical and likely reflects a data coding artefact in one of the source datasets rather than genuine security incidents.

5.4 Analysis 4 — Delay Cause Share per Airline

Visualisation: Stacked 100% bar chart — one bar per airline, segments showing the percentage share of each delay cause.

Selected results (% of total delay minutes):

Airline	Carrier	Weather	NAS	Security	Late A/C
Hawaiian Airlines	56.8	3.3	2.2	0.2	37.4
Delta Air Lines	40.0	7.5	23.3	0.1	29.0
Skywest Airlines	39.2	7.4	16.2	0.2	37.0
Southwest Airlines	31.8	3.8	14.0	0.2	50.3
Virgin America	19.3	5.5	36.7	0.3	38.2
Spirit Air Lines	24.9	3.3	39.8	0.4	31.6

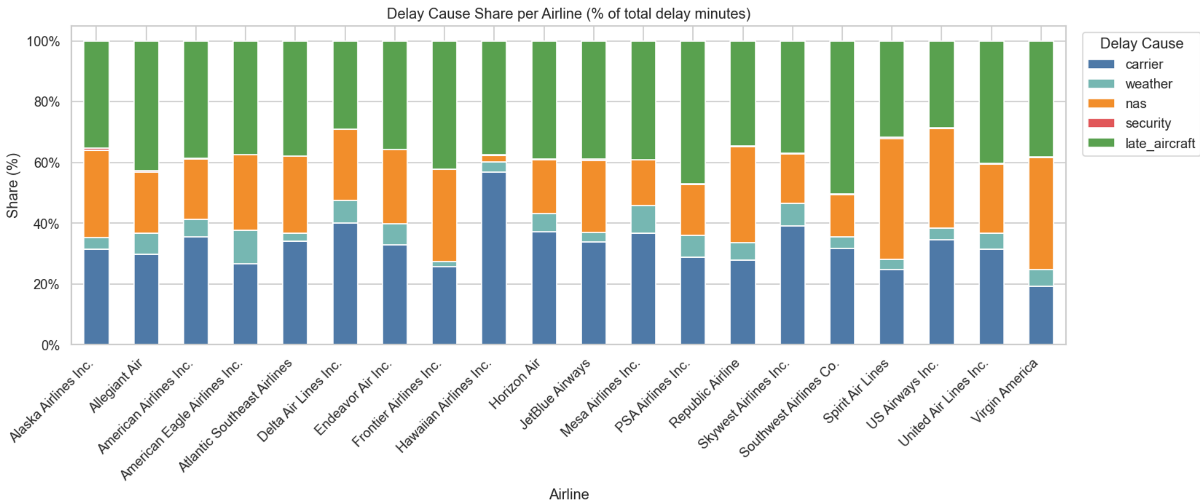

Figure 4: Delay cause share per airline (% of total delay minutes, stacked 100% bar).

Key observations:

- **Late inbound aircraft** is the largest single delay cause for Southwest (50.3%) — consistent with its high-frequency, point-to-point network where a delay on one leg cascades forward.
- **Hawaiian Airlines** shows an unusually high carrier delay share (56.8%), with almost no NAS delay (2.2%), which reflects its primarily over-ocean routes where ATC congestion is minimal.
- **NAS delay** dominates for Spirit (39.8%) and Virgin America (36.7%), suggesting heavy reliance on congested hub airports where ATC flow management causes significant holds.
- Carrier delay + late-aircraft together account for over 60% of total delay minutes for every airline in the dataset, confirming that airline-controllable factors are the dominant source of delay.

5.5 Analysis 5 — Weekend vs Weekday Delays by Quarter

Visualisation: Two-panel grouped bar chart — left panel shows percentage of significantly-delayed flights (>15 min); right panel shows average departure delay. Both panels group quarters on the x-axis with weekend/weekday side by side.

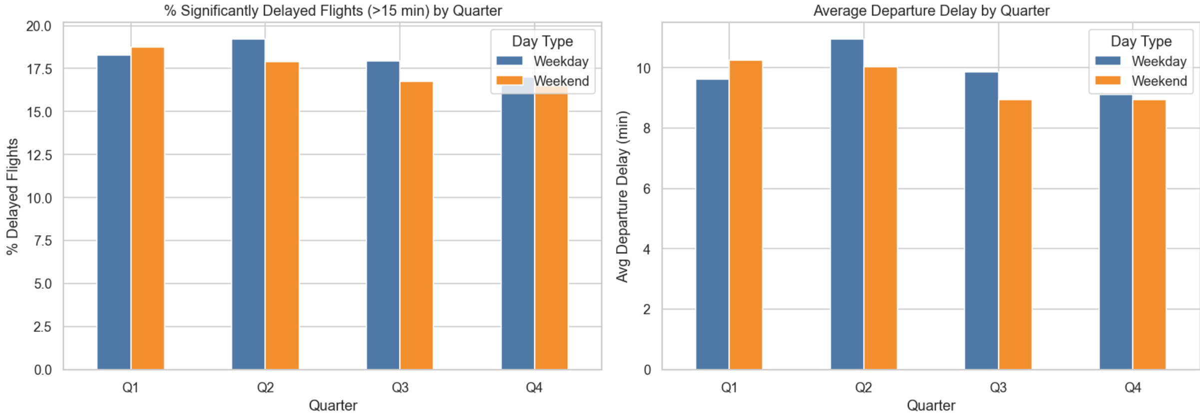

Figure 5: Percentage of significantly delayed flights and average departure delay by quarter and day type.

Full result set:

Quarter	Day type	Flights	Sig. delayed	Delayed %	Avg delay (min)
Q1	Weekday	1 563 139	286 003	18.30	9.63
Q1	Weekend	564 132	105 698	18.74	10.24
Q2	Weekday	1 614 570	310 175	19.21	10.95
Q2	Weekend	579 159	103 664	17.90	10.04
Q3	Weekday	1 646 533	295 138	17.92	9.87
Q3	Weekend	590 318	98 887	16.75	8.94
Q4	Weekday	1 135 719	193 090	17.00	9.12
Q4	Weekend	421 618	69 593	16.51	8.95

Key observations:

- The weekend vs weekday effect is **small and inconsistent**. In Q1, weekends are marginally worse (18.74% vs 18.30%); in Q2, Q3, and Q4, weekdays are actually worse.
- **Q2 weekdays** have the highest significantly-delayed rate (19.21%) and average delay (10.95 min), likely driven by spring thunderstorm season.
- The hypothesis that weekends are systematically worse is **not supported** by the data. Seasonal effects (quarter) explain more variance than the weekend flag.

5.6 Analysis 6 — Top 20 Airports: Volume, Delay and Cancellation Rate

Visualisation: Scatter plot — x-axis = outbound flight count, y-axis = average departure delay, bubble size proportional to total cancellations, colour indicating cancellation rate (%).

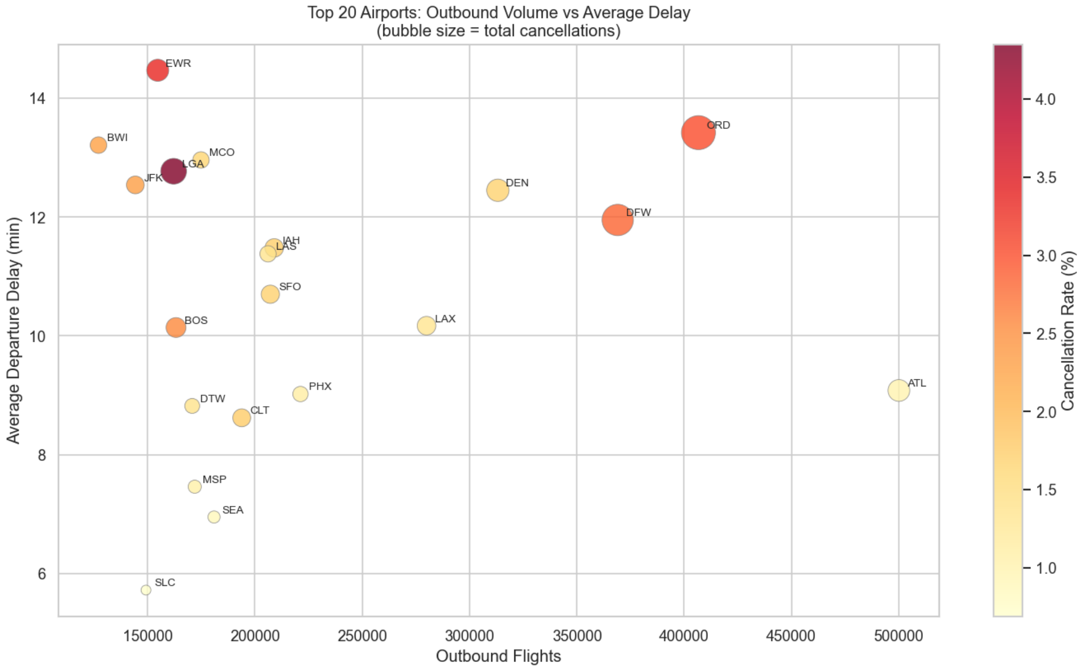

Figure 6: Top 20 airports: outbound volume vs average departure delay. Bubble size = total cancellations; colour = cancellation rate (%).

Full result set:

IATA	City	Flights	Avg delay	Cancellations	Cancel %
ATL	Atlanta	500 076	9.08	5 015	1.00
ORD	Chicago	406 670	13.42	12 265	3.02
DFW	Dallas-Fort Worth	369 057	11.95	10 430	2.83
DEN	Denver	313 203	12.45	5 297	1.69
LAX	Los Angeles	279 977	10.17	3 683	1.32
PHX	Phoenix	221 226	9.02	2 494	1.13
IAH	Houston	209 010	11.48	3 613	1.73
SFO	San Francisco	207 184	10.70	3 525	1.70
LAS	Las Vegas	206 114	11.38	2 807	1.36
CLT	Charlotte	193 856	8.62	3 384	1.75
SEA	Seattle	180 949	6.95	1 577	0.87
MCO	Orlando	174 859	12.96	2 859	1.64
MSP	Minneapolis	171 955	7.46	1 834	1.07
DTW	Detroit	170 779	8.82	2 360	1.38
BOS	Boston	163 234	10.14	4 180	2.56
LGA	New York (LGA)	162 128	12.77	7 051	4.35
EWR	Newark	154 718	14.47	5 161	3.34
SLC	Salt Lake City	149 237	5.72	1 027	0.69
JFK	New York (JFK)	144 277	12.54	3 339	2.31
BWI	Baltimore	127 109	13.21	2 912	2.29

Key observations:

- **Atlanta (ATL)** is by far the busiest airport (500 076 outbound flights) yet has a below-average delay (9.08 min) and the lowest cancellation rate among the top-5

(1.00%), demonstrating that high traffic volume does not necessarily imply poor performance.

- **LaGuardia (LGA)** and **Newark (EWR)** stand out with the highest cancellation rates (4.35% and 3.34% respectively) and among the highest average delays (12.77 and 14.47 min), consistent with their known congestion problems in the New York metro airspace.
- **Salt Lake City (SLC)** and **Seattle (SEA)** are the best-performing large airports: SLC averages only 5.72 min departure delay with a 0.69% cancellation rate; SEA averages 6.95 min with 0.87%.
- **Chicago O'Hare (ORD)** and **Dallas-Fort Worth (DFW)** combine very high traffic (>360 000 flights each) with elevated cancellation rates (>2.8%), confirming their reputation as weather- and congestion-prone hubs.